

Retail Sales Analytics Project Report

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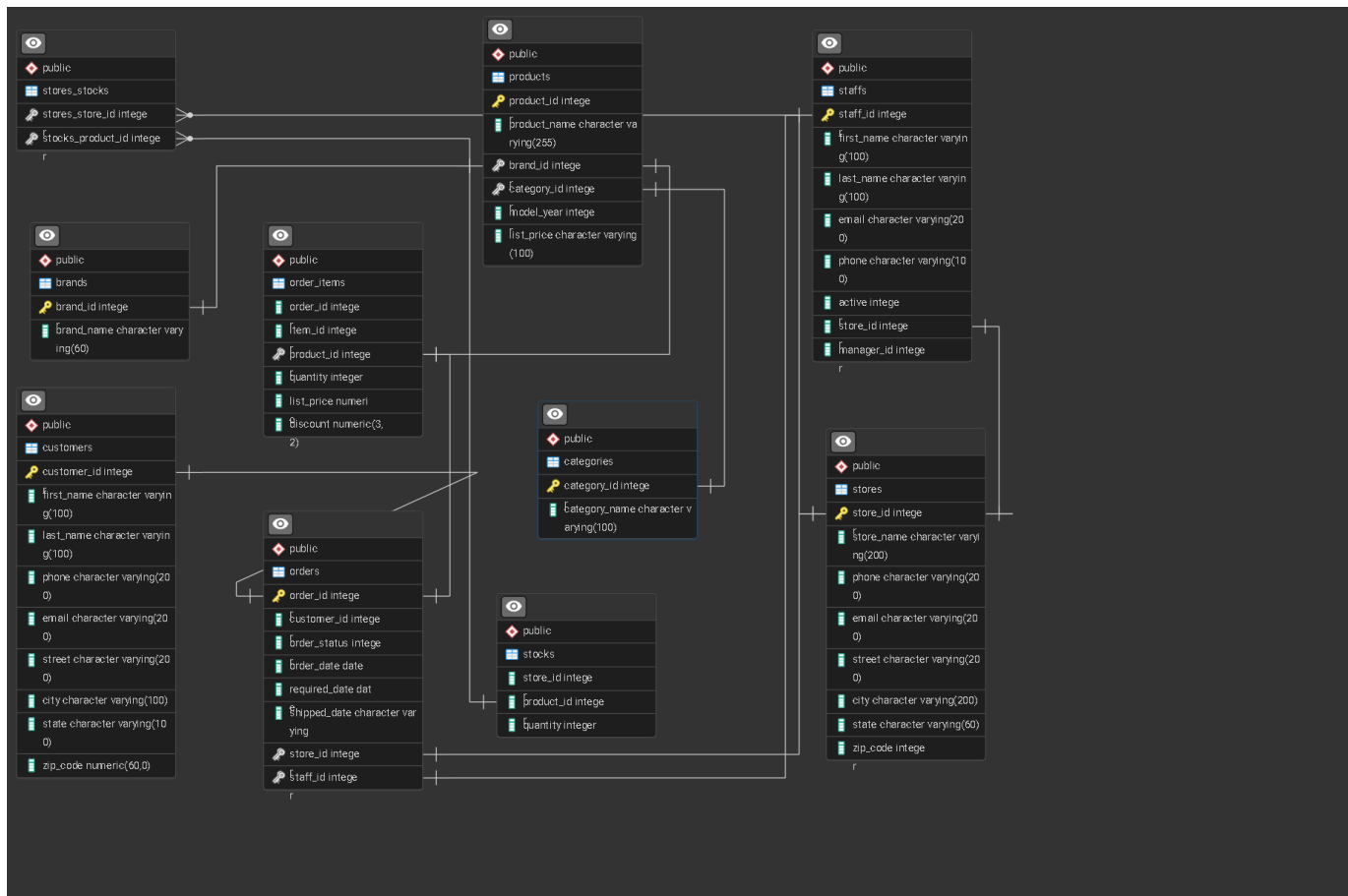
1. Executive Summary

This project integrates **PostgreSQL** (for data storage and queries) with **Power BI** (for visualization) to deliver actionable insights across sales, inventory, customer behavior, and staff performance.

The solution enables management to monitor KPIs, optimize inventory, evaluate staff, and improve customer satisfaction.

2. Data Model & ER Diagram

- **Entities:** Customers, Orders, Order_Items, Products, Brands, Categories, Stores, Staffs, Stocks.
- **Relationships:**
 - Customers → Orders (1:N)
 - Orders → Order_Items (1:N)
 - Order_Items → Products (N:1)
 - Products → Brands/Categories (N:1)
 - Orders → Stores (N:1)
 - Orders → Staffs (N:1)
 - Stores → Stocks → Products (M:N)



1. **Identify top-selling brands by region and store** → Marketing strategy alignment.
2. **Evaluate staff performance based on total sales handled** → HR performance reviews.
3. **Track customer orders and fulfillment status** → Improve delivery reliability.
4. **Identify the most profitable product categories** → Guide product expansion.
5. **Analyze stock levels across stores to optimize inventory** → Prevent stockouts/overstocking.
6. **Understand order trends across time & monitor delayed shipments** → Forecasting & logistics.
7. **Discover customer concentration and demographics** → Targeted marketing campaigns.

- 1. Strengthen Regional Brand Strategy**
 - Focus promotions on top-selling brands by region/store.
 - Negotiate supplier contracts for high-performing brands.

2. Enhance Staff Productivity

- Reward top performers, train low performers.
- Align staff allocation with high-traffic stores.

3. Improve Customer Experience

- Monitor fulfillment status to reduce delays.
- Communicate proactively with customers on pending orders.
- Launch loyalty programs for repeat customers.

4. Optimize Product Portfolio

- Invest in profitable categories, phase out weak ones.
- Use insights to guide new product launches.

5. Inventory Management & Cost Control

- Balance stock levels across stores.
- Automate reorder alerts for low inventory.
- Negotiate bulk discounts with suppliers.

6. Forecasting & Logistics

- Use order trends for demand forecasting.
- Monitor delayed shipments vs. required dates.
- Collaborate with logistics partners for timely delivery.

7. Customer Segmentation & Targeted Marketing

- Identify concentration/demographics for campaigns.
- Focus retention on high-value customers.
- Expand into untapped regions.

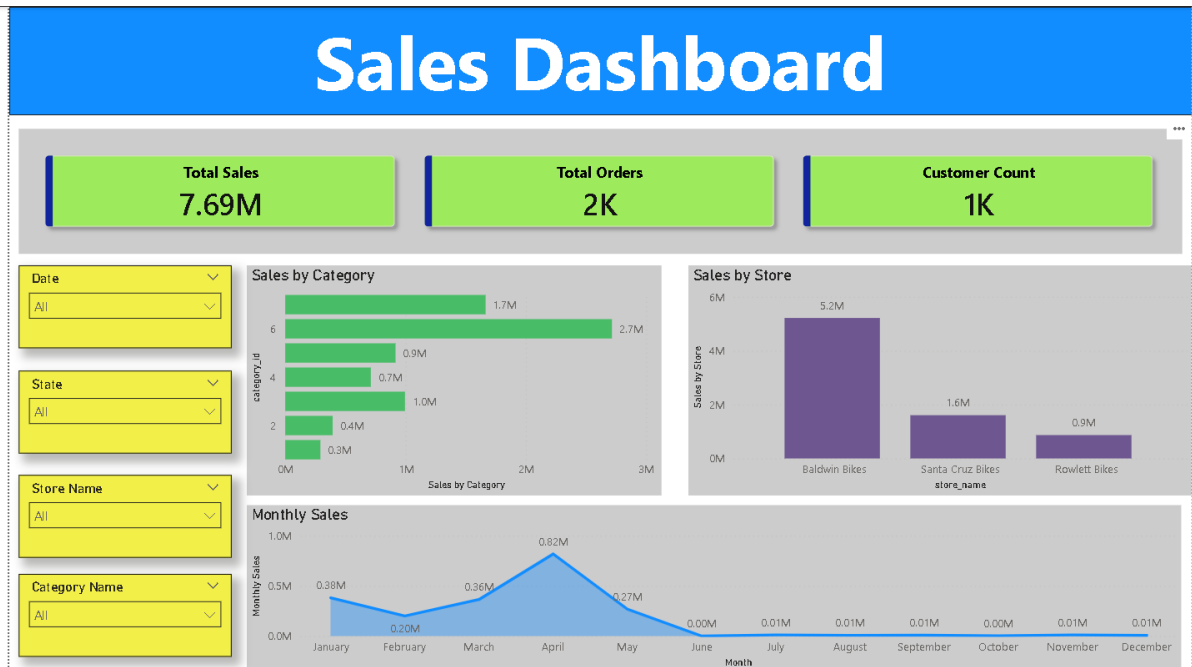
5. Technical Implementation

- **SQL Views:** Created for KPIs (sales, inventory, staff, customers).
- **DAX Measures:** Defined for Power BI (Total Sales, Monthly Sales, Stock Levels, Customer Sales, Staff Sales).
- **Power BI Features:** Drill-down, slicers, conditional formatting, interactive maps.

6. Dashboards Overview

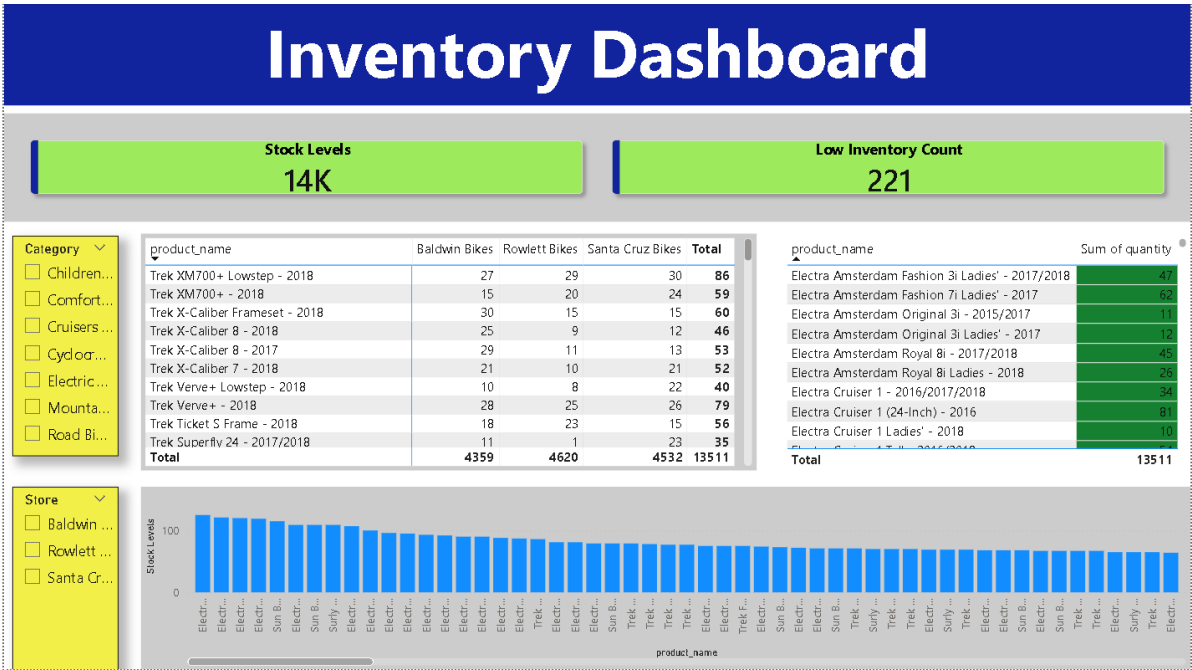
Dashboard 1 – Sales

- KPIs: Total Sales, Total Orders, Customer Count
- Bar Charts: Sales by Category, Sales by Store
- Line Chart: Monthly Sales Trend
- Slicers: Date, Region, Store, Category



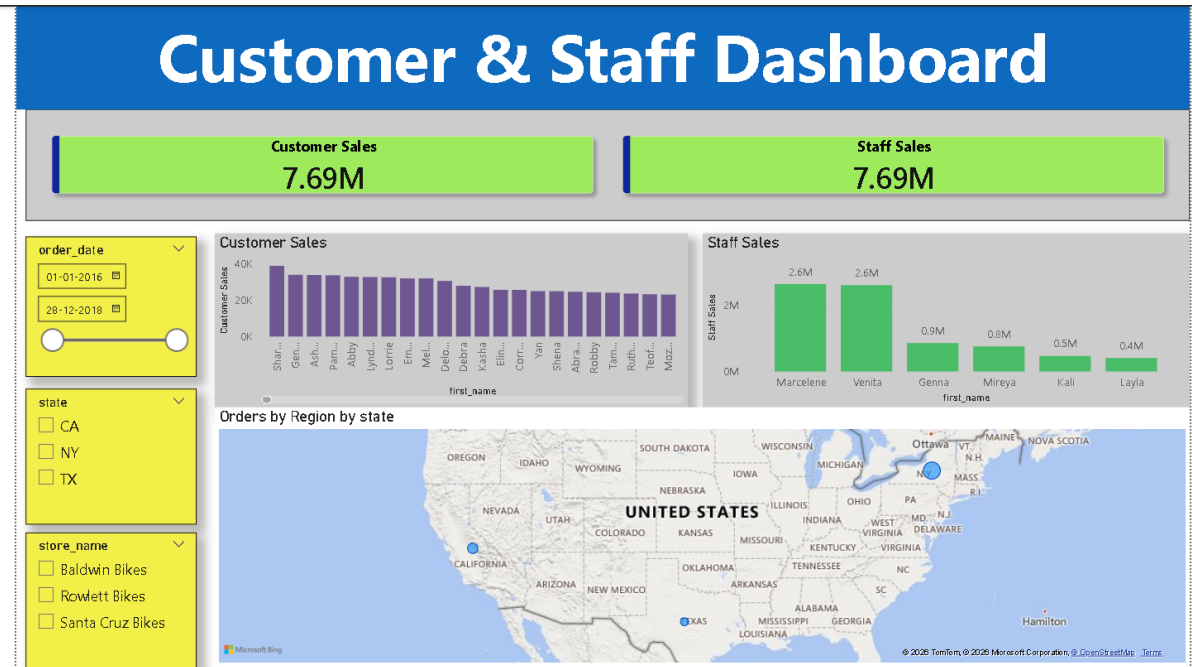
Dashboard 2 – Inventory

- KPIs: Total Stock Levels, Low Inventory Count
- Bar Chart: Stock Levels by Product
- Table: Low Inventory Products (conditional formatting)
- Matrix: Product Availability by Store
- Slicers: Category, Store



Dashboard 3 – Customer & Staff

- KPIs: Top Customer Sales, Staff Sales Performance
- Bar Charts: Top Customers, Staff Performance
- Map: Orders by Region
- Slicers: Date, Region, Store



7. Deliverables

- SQL scripts for analysis queries.

```
SELECT * FROM order_items;
```

```
ALTER TABLE order_items
```

```
ALTER COLUMN list_price TYPE NUMERIC
```

```
USING CAST(REPLACE(REPLACE(TRIM(list_price), '₹', ''), ',', '' ) AS NUMERIC);
```

```
-- Store-wise total sales revenue
```

```
SELECT s.store_name,
```

```
       SUM(oi.quantity * oi.list_price * (1 - oi.discount)) AS total_sales
```

```
FROM orders o
```

```
JOIN order_items oi ON o.order_id = oi.order_id
```

```
JOIN stores s ON o.store_id = s.store_id
```

```
GROUP BY s.store_name
```

```
ORDER BY total_sales DESC;
```

```
-- Region-wise (state) sales revenue
```

```
SELECT s.state,
```

```
       SUM(oi.quantity * oi.list_price * (1 - oi.discount)) AS total_sales
```

```
FROM orders o
```

```
JOIN order_items oi ON o.order_id = oi.order_id
```

```
JOIN stores s ON o.store_id = s.store_id
```

```
GROUP BY s.state
```

```
ORDER BY total_sales DESC;
```

```
-- Product-wise sales quantity and revenue
```

```
SELECT p.product_name,
```

```
       SUM(oi.quantity) AS total_units_sold,
```

```
       SUM(oi.quantity * oi.list_price * (1 - oi.discount)) AS revenue
```

```
FROM order_items oi
```

```
JOIN products p ON oi.product_id = p.product_id
```

```
GROUP BY p.product_name
```

```
ORDER BY revenue DESC;
```

```
-- Product inventory vs sales trend
```

```
SELECT p.product_name,  
       SUM(oi.quantity) AS units_sold,  
       SUM(st.quantity) AS current_stock  
FROM products p  
JOIN order_items oi ON p.product_id = oi.product_id  
JOIN stocks st ON p.product_id = st.product_id  
GROUP BY p.product_name  
ORDER BY units_sold DESC;
```

```
-- Staff-wise sales handled
```

```
SELECT st.first_name || ' ' || st.last_name AS staff_name,  
       COUNT(DISTINCT o.order_id) AS orders_handled,  
       SUM(oi.quantity * oi.list_price * (1 - oi.discount)) AS total_sales  
FROM staffs st  
JOIN orders o ON st.staff_id = o.staff_id  
JOIN order_items oi ON o.order_id = oi.order_id  
GROUP BY staff_name  
ORDER BY total_sales DESC;
```

```
-- Customer order frequency and spending
```

```
SELECT c.first_name || ' ' || c.last_name AS customer_name,  
       COUNT(DISTINCT o.order_id) AS total_orders,  
       SUM(oi.quantity * oi.list_price * (1 - oi.discount)) AS total_spent  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
JOIN order_items oi ON o.order_id = oi.order_id  
GROUP BY customer_name  
ORDER BY total_orders DESC;
```

-- Top repeat customers

```
SELECT c.first_name || ' ' || c.last_name AS customer_name,  
       COUNT(o.order_id) AS order_count  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
GROUP BY customer_name  
HAVING COUNT(o.order_id) > 5  
ORDER BY order_count DESC;
```

-- Average order value per customer

```
SELECT c.first_name || ' ' || c.last_name AS customer_name,  
       AVG(oi.quantity * oi.list_price * (1 - oi.discount)) AS avg_order_value  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
JOIN order_items oi ON o.order_id = oi.order_id  
GROUP BY customer_name  
ORDER BY avg_order_value DESC;
```

-- Overall revenue and discount impact

```
SELECT SUM(oi.quantity * oi.list_price) AS gross_revenue,  
       SUM(oi.quantity * oi.list_price * oi.discount) AS total_discount,  
       SUM(oi.quantity * oi.list_price * (1 - oi.discount)) AS net_revenue  
FROM order_items oi;
```

-- Discount effectiveness by product

```
SELECT p.product_name,  
       SUM(oi.quantity * oi.list_price * oi.discount) AS discount_given,  
       SUM(oi.quantity * oi.list_price * (1 - oi.discount)) AS net_revenue  
FROM order_items oi  
JOIN products p ON oi.product_id = p.product_id
```

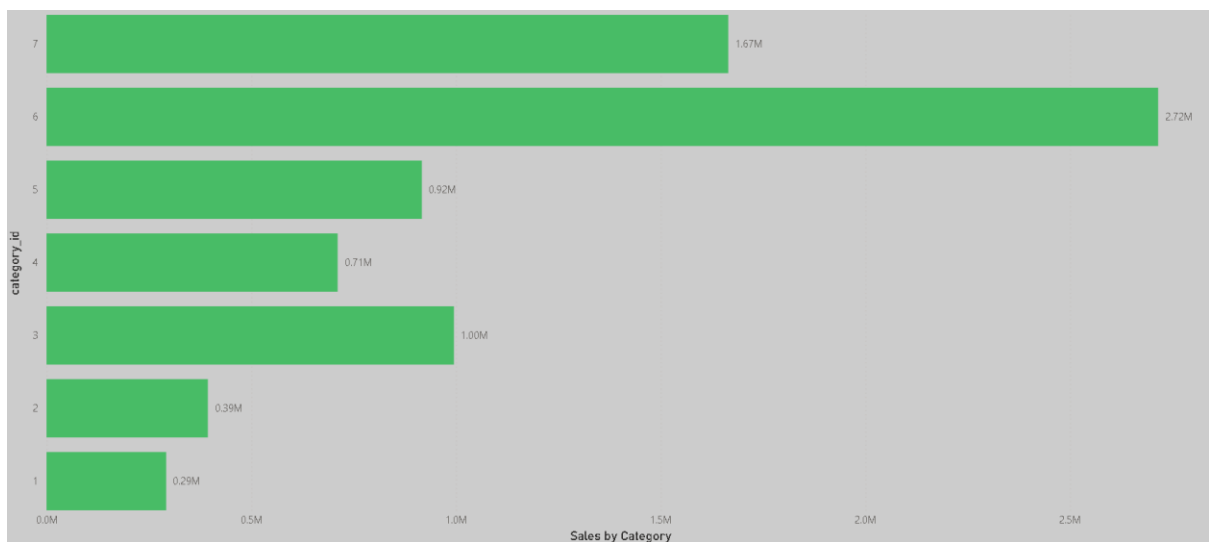

GROUP BY p.product_name

ORDER BY discount_given DESC;

- **Power BI dashboards (Sales, Inventory, Customer & Staff).**

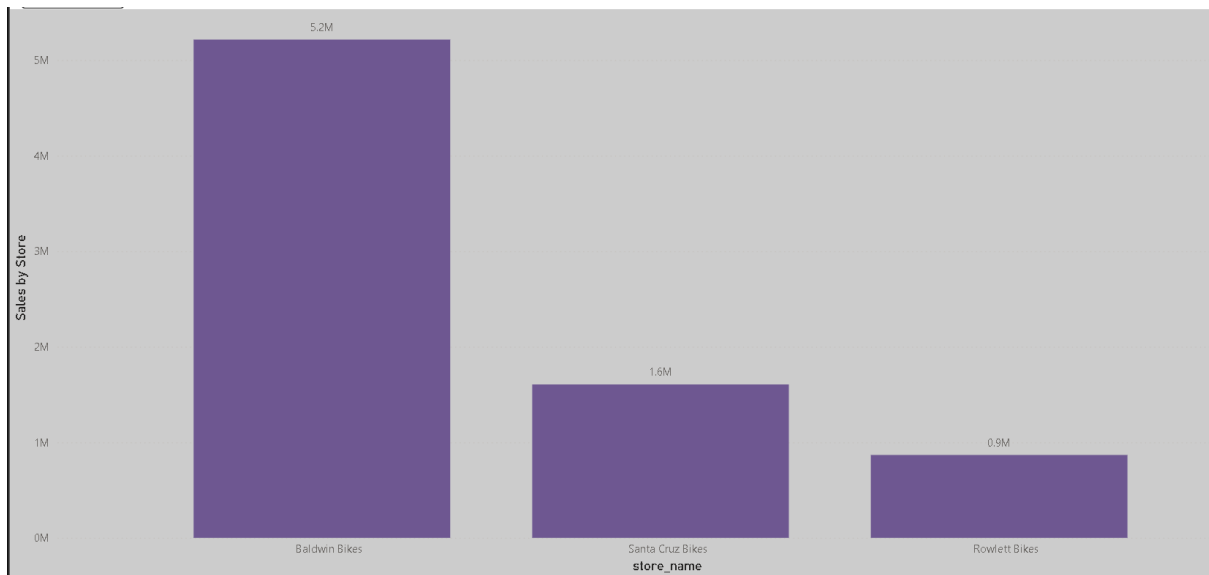
Sales by Category (Bar Chart)

- One category dominates with **2.7M sales**, far ahead of others.
- Mid-range categories contribute ~0.9M–1.7M, while smaller ones are <0.5M.
- Insight: Focus marketing on top category, but explore growth in mid-range categories.



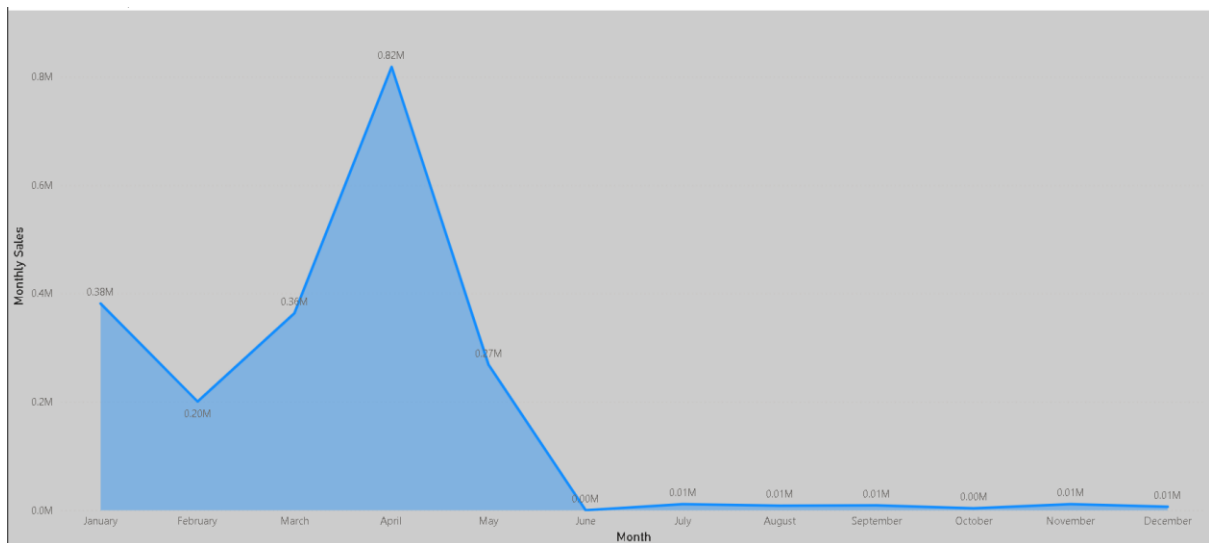
Sales by Store (Bar Chart)

- **Baldwin Bikes** leads with **5.2M sales**, ~68% of total.
- Santa Cruz (1.6M) and Rowlett (0.9M) lag significantly.
- Insight: Baldwin is the flagship store; other stores need targeted promotions.



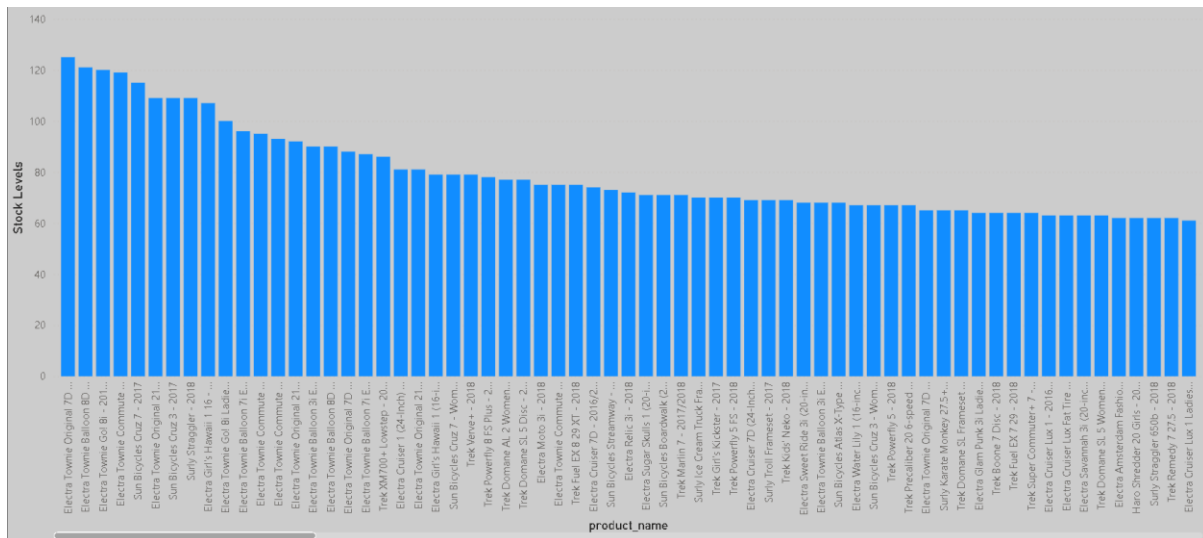
Monthly Sales (Line Chart)

- Strong peaks in **April (0.82M)** and **January (0.5M)**.
- Weak months include February (0.2M) and May (0.27M).
- Insight: Seasonality exists — plan campaigns around high-performing months.



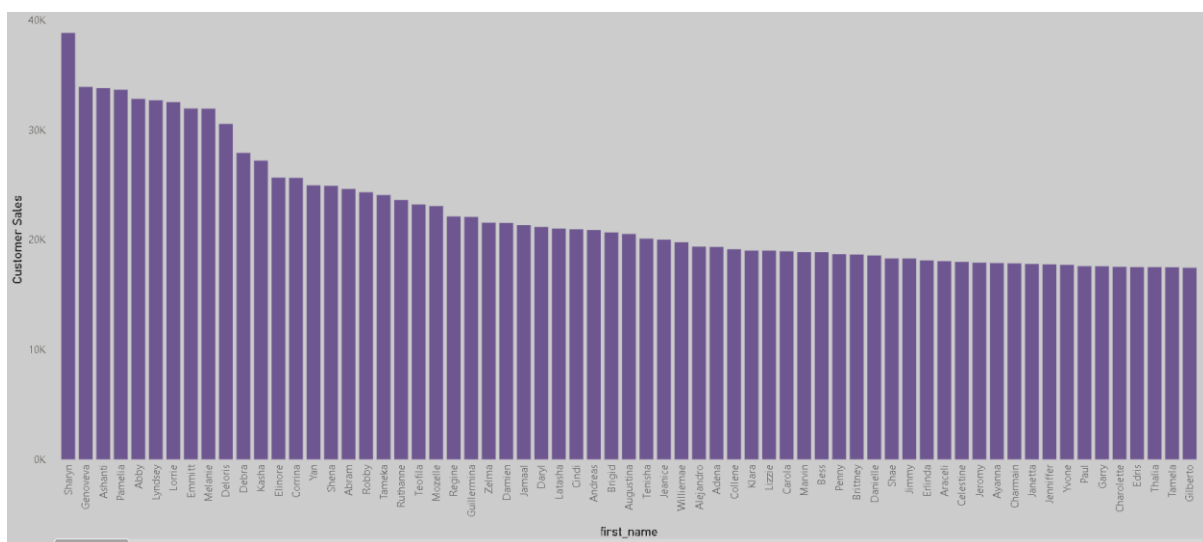
Stock Levels by Product (Bar Chart)

- Certain products dominate inventory, while many have minimal stock.
- Insight: Overstock risk for high-quantity items; balance distribution.



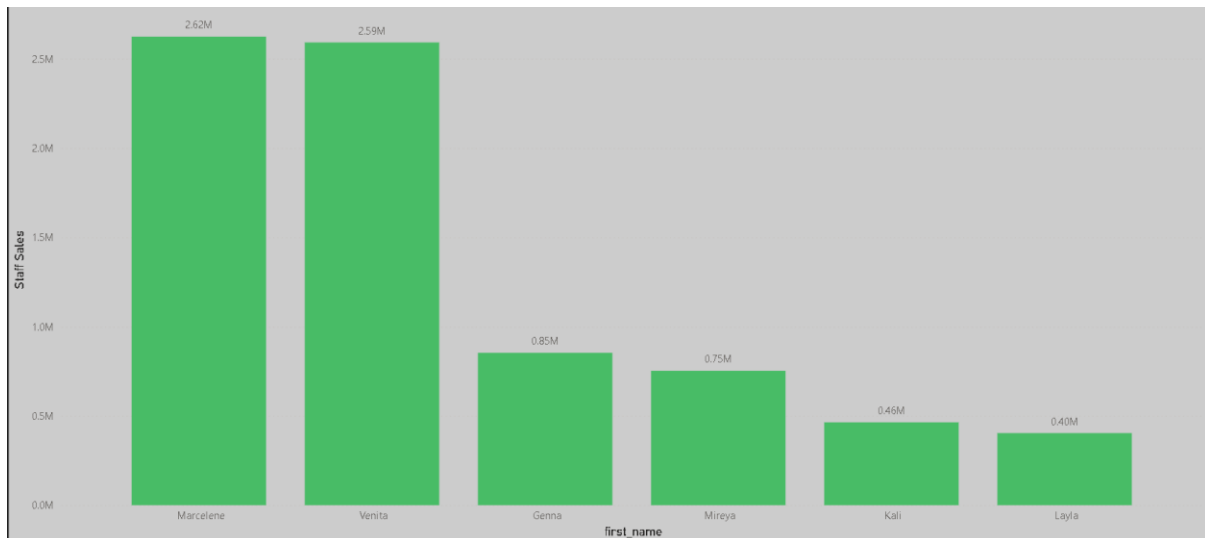
Top Customers (Bar Chart)

- A handful of customers (Shari, Cinn., Ash., Pam., Abby) contribute significantly.
- Insight: These are **high-value customers** — target with loyalty programs.



Staff Performance (Bar Chart)

- Staff like **Marcelene (2.6M)** and **Venita (2.2M)** outperform others.
- Insight: Recognize top performers, train/support lower performers.



8. Conclusion

This project demonstrates how integrating **PostgreSQL** with **Power BI** can transform raw transactional data into actionable business intelligence.

- The **Sales Dashboard** highlights revenue drivers, top categories, and store performance, enabling management to focus on profitable segments and seasonal demand.
- The **Inventory Dashboard** provides visibility into stock distribution and low-inventory alerts, helping prevent stockouts and optimize supply chain efficiency.
- The **Customer & Staff Dashboard** uncovers high-value customers, evaluates staff contributions, and maps regional demand, supporting both customer retention and workforce management.

From the analysis:

- **Sales are concentrated** in one store and one category, suggesting both strength and risk of over-dependence.
- **Inventory imbalances** across stores highlight the need for redistribution and smarter replenishment.
- **Customer loyalty** is driven by a small group of repeat buyers, while **staff performance** varies significantly, offering opportunities for recognition and training.
- **Regional concentration** in California, Texas, and New York points to untapped markets elsewhere.

The **business recommendations** derived from these insights — strengthening brand strategy, enhancing staff productivity, improving customer experience, optimizing product portfolios, balancing inventory, forecasting demand, and targeting marketing campaigns — provide a roadmap for sustainable growth.